

OCCUPATIONAL INJURY RISK ANALYSIS

Quebec Workforce | 2022 vs. 2023

Based on CNESST Open Data — 276,307 Injury Claims

Prepared as: Data Analytics Work Sample

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Note: All data is publicly available. No confidential information is included.

1. Executive Summary

This report analyzes 276,307 occupational injury claims filed in Quebec between 2022 and 2023, drawn from the CNESST open data portal. The analysis identifies key risk patterns across industry sectors, injury types, demographic groups, and emerging risk categories including psychological injuries and musculoskeletal disorders. Findings are intended to support risk-informed decision-making for organizational leadership.

Key Performance Indicators at a Glance

161,962 Total Claims (2022) Baseline Year	114,345 Total Claims (2023) ▼ -29.4% YoY	41.5% to 16.3% COVID-19 Claims Drop ▼ -25.2 pts	2.6% to 4.8% Psychological Injuries ▲ +2.2 pts
Health Care Top Risk Sector 30.6% of claims	Overexertion Highest Incident Type #1 in 2023	16.6% to 23.3% Musculoskeletal Flag ▲ +6.7 pts	3,729 to 4,786 Assault/Violence Claims ▲ +28.4% YoY

Headline Finding

The steep overall decline in claims is largely explained by the withdrawal of COVID-19-related injuries from 2022 to 2023. When COVID claims are removed, the underlying injury burden remains significant and several risk categories, including psychological injuries, musculoskeletal disorders, and workplace violence, are trending upward.

Three Priority Findings

- Physical and chronic injury risks are rising: musculoskeletal disorders increased from 16.6% to 23.3% of all claims, signalling growing strain from repetitive work environments.
- Workplace violence is escalating: assault and violence claims grew 28.4% year-over-year, concentrated in Health Care and Education sectors.
- Psychological injury is an emerging systemic risk: claims flagged as psychological nearly doubled in share (2.6% to 4.8%), with Health Care workers accounting for 42.5% of all 2023 psychological claims.

2. Claim Volume & Demographic Profile

2.1 Overall Claim Volume

Total occupational injury claims declined from 161,962 in 2022 to 114,345 in 2023, a year-over-year reduction of 47,617 claims (-29.4%). This reduction is primarily attributable to the post-pandemic normalization of COVID-19-related claims, which fell from 67,202 (41.5% of 2022 claims) to 18,662 (16.3% of 2023 claims). Excluding COVID-19 flagged claims, the underlying injury volume declined more modestly from approximately 94,760 in 2022 to 95,683 in 2023, effectively flat, indicating that core occupational risk has not meaningfully diminished.

Metric	2022	2023	Change
Total claims	161,962	114,345	-29.4%
COVID-19 flagged claims	67,202 (41.5%)	18,662 (16.3%)	-72.2%
Non-COVID claims (est.)	94,760	95,683	+0.97%
Musculoskeletal flagged	26,814 (16.6%)	26,624 (23.3%)	+6.7 pts
Psychological flagged	4,190 (2.6%)	5,532 (4.8%)	+2.2 pts
Machine injury flagged	3,930 (2.4%)	4,605 (4.0%)	+1.6 pts
Hearing loss flagged	11,350 (7.0%)	9,789 (8.6%)	+1.6 pts

2.2 Sex Distribution

In 2022, female workers accounted for the majority of claims (52.4% vs. 47.6% male), driven heavily by the Health Care and Social Assistance sector where women represent 81.6% of injured workers. In 2023, the sex distribution shifted slightly, with male claimants rising to 55.0% — a reversal largely explained by the proportional decline of Health Care COVID-19 claims, which disproportionately affected female workers.

Sex	2022 Claims	2022 Share	2023 Claims	2023 Share
Female	84,802	52.4%	51,428	45.0%
Male	77,160	47.6%	62,917	55.0%

Risk Insight

The high proportion of female claimants in Health Care reflects both workforce composition and sectoral risk concentration. As the sector with the highest psychological and violence-related claims, targeted gender-informed interventions are warranted.

2.3 Age Group Distribution

The highest-risk age cohorts are consistently the 35-44 and 45-54 groups, together accounting for approximately 43-44% of all claims in both years. The 55-64 cohort is growing in share (18.2% to 19.3%), reflecting both an aging workforce and potential cumulative exposure effects. Young workers (under 20) remain stable at 2.4% but represent a disproportionate risk per hours worked given their typically entry-level roles in high-risk sectors.

Age Group	2022 Claims	2022 Share	2023 Claims	2023 Share
Under 20	3,921	2.4%	2,759	2.4%
20-24	12,924	8.0%	8,213	7.2%
25-34	34,095	21.1%	22,652	19.8%
35-44 (Peak)	36,683	22.6%	24,840	21.7%
45-54 (Peak)	35,194	21.7%	24,856	21.7%
55-64	29,431	18.2%	22,071	19.3%
65 and Over	9,714	6.0%	8,954	7.8%

3. Industry Sector Risk Analysis

3.1 Sector Overview

Health Care and Social Assistance dominates the injury landscape in both years, accounting for 46.0% of claims in 2022 and 30.6% in 2023. While the absolute number declined sharply (74,517 to 34,944, or -53.1%), this is primarily attributable to COVID-19 claim normalization rather than genuine safety improvement. The sector continues to carry the highest burden of psychological injuries, musculoskeletal disorders, and workplace violence of any sector in the dataset.

Manufacturing and Construction sectors held steady in absolute terms, making their relative share of risk more prominent as COVID-19 claims exit the denominator. This shift demands renewed attention to physical safety in manual labour environments.

Industry Sector	2022 Claims	2023 Claims	Change	2023 Share
Health Care & Social Assistance	74,517	34,944	-53.1%	30.6%
Durable Goods Manufacturing	10,297	10,547	+2.4%	9.2%
Construction	9,502	9,590	+0.9%	8.4%
Non-Durable Goods Mfg.	7,883	7,540	-4.4%	6.6%
Retail Trade	7,216	7,266	+0.7%	6.4%
Educational Services	5,625	5,803	+3.2%	5.1%
Transportation & Warehousing	6,193	5,739	-7.3%	5.0%
Public Administration	5,728	5,155	-10.0%	4.5%
Business & Building Support	5,565	4,545	-18.3%	4.0%

Sector Risk Flag — Education

Educational Services saw a +3.2% increase in claims and a 52.8% increase in assault/violence incidents (677 to 1,035). This sector warrants close monitoring as an emerging violence risk environment.

3.2 Health Care Sector Deep Dive

Health Care remains the single highest-risk sector across every major risk dimension. Within this sector in 2023:

- 81.6% of injured workers are female, reflecting workforce composition.
- The 35-54 age band accounts for 51.2% of sector claims.
- The leading incident type is Exposure to Caustic/Harmful/Allergenic Substances (18,501 claims), driven by residual COVID-19 protocol exposures.
- Overexertion (5,706 claims) is the second-highest incident type — indicating physical workload risk independent of the pandemic.
- Assault/Violence by Person accounts for 2,173 claims — the highest of any sector and growing year-over-year.
- 42.5% of all psychological injury claims in 2023 originate from this sector.
- 26.9% of all musculoskeletal claims in 2023 originate from this sector.

Strategic Implication

The Health Care sector faces a compounding risk profile: physical overexertion, chemical exposure, workplace violence, and psychological injury are all present and growing. A sector-specific risk mitigation strategy — addressing both physical and psychosocial hazards — would have the greatest impact on the overall claims portfolio.

4. Injury Profile Analysis

4.1 Nature of Injury

Traumatic Muscle and Tendon Injuries emerged as the leading injury type in 2023 (30.8% of claims), overtaking Symptoms/III-Defined Conditions which dominated in 2022 (30.5%) due to COVID-19 presentation patterns. This shift marks a meaningful return to physical trauma as the primary injury driver — an important signal for prevention programming focused on physical safety protocols.

Injury Nature	2022 Claims	2022 Share	2023 Claims	2023 Share
Traumatic Muscle/Tendon Injury	37,351	23.1%	35,188	30.8%
Symptoms/III-Defined Conditions	49,435	30.5%	14,670	12.8%
Wound/Superficial Contusion	9,783	6.0%	11,415	10.0%
Nervous System/Sensory Disorder	12,049	7.4%	10,040	8.8%
Musculoskeletal Disorder	8,009	4.9%	8,487	7.4%
Traumatic Bone/Nerve/Spinal	8,794	5.4%	7,690	6.7%
Respiratory Disease	18,086	11.2%	4,487	3.9%
Mental Disorder/Syndrome	1,686	1.0%	N/A	—

4.2 Incident Type (How Injuries Occur)

Overexertion rose to the second most common incident type in 2023 (17.0%) and is poised to become the leading category as COVID-19 exposures decline further. Falls on the Same Level (10.0%) remain a persistent hazard. The most significant trend is the rise in Assault/Violence by Person — up from 2.3% to 4.2% of all claims, representing a 28.4% increase in absolute claim count.

Incident Type	2022 Claims	2022 Share	2023 Claims	2023 Share
Exposure to Caustic/Harmful Substance	69,164	42.7%	21,002	18.4%
Overexertion	16,990	10.5%	19,431	17.0%
Bodily Reaction	13,808	8.5%	12,507	10.9%
Fall on Same Level	11,054	6.8%	11,382	10.0%
Struck by Object	10,593	6.5%	9,891	8.7%
Noise Exposure	11,323	7.0%	9,289	8.1%
Assault/Violence by Person	3,729	2.3%	4,786	4.2%
Fall to Lower Level	4,919	3.0%	4,787	4.2%

Violence Risk — Critical Trend

Assault and violence claims grew from 3,729 to 4,786 — a 28.4% increase. Health Care (2,173), Educational Services (1,035), and Public Administration (323) are the top three affected sectors. This trend is particularly relevant for organizations with public-facing or care-delivery roles.

4.3 Causal Agents (What Caused the Injury)

The causal agent landscape shifted dramatically between years. Infectious and Parasitic Agents dominated in 2022 (41.6%) due to COVID-19 but fell to 16.4% in 2023. The worker's own body (Person — Worker) became the leading cause in 2023 at 16.7%, consistent with overexertion being the primary incident type. Floors and Ground Surfaces remained a persistent top-three causal agent, rising from 8.4% to 12.8%.

Causal Agent	2022 Claims	2022 Share	2023 Claims	2023 Share
Person (Worker) — overexertion/self	21,864	13.5%	19,129	16.7%
Infectious & Parasitic Agents	67,375	41.6%	18,713	16.4%
Floors/Walkways/Ground Surfaces	13,552	8.4%	14,634	12.8%
Person (Non-Worker) — violence	8,260	5.1%	10,095	8.8%
Atmospheric/Environmental Conditions	11,848	7.3%	9,879	8.6%
Atmospheric Pressure Containers	4,425	2.7%	5,197	4.5%
Construction Materials (Solid)	4,728	2.9%	4,551	4.0%

5. Emerging Risk Categories

5.1 Psychological Injuries

Psychological injuries represent the fastest-growing risk category in the dataset. The share of claims flagged as psychological nearly doubled from 2.6% in 2022 (4,190 claims) to 4.8% in 2023 (5,532 claims). This growth occurred even as total claims declined sharply — meaning psychological risk is intensifying on an absolute and relative basis simultaneously.

Health Care and Social Assistance accounts for 42.5% of all psychological claims in 2023, followed by Educational Services (19.4%) and Public Administration (8.4%). The concentration of psychological claims in people-facing, high-stress service sectors points to systemic workload and interpersonal risk factors rather than isolated incidents.

Sector	2023 Psychological Claims	% of All Psych Claims
Health Care & Social Assistance	2,352	42.5%
Educational Services	1,075	19.4%
Public Administration	465	8.4%
Transportation & Warehousing	397	7.2%
Business & Building Support Services	280	5.1%
Other / Not Coded	191	3.5%
All other sectors	772	13.9%

Strategic Priority

Psychological injury is no longer a marginal risk category. Its 84.6% growth in claim share over one year, concentrated in care and public-service sectors, signals that workload stress, trauma exposure, and workplace violence are generating measurable harm at scale. Early intervention programs and psychosocial risk assessments in top affected sectors are warranted.

5.2 Musculoskeletal Disorders

Musculoskeletal injuries, the category of chronic, repetitive-strain injuries affecting muscles, joints, and connective tissue, represent the largest special risk category in the 2023 dataset, with 26,624 flagged claims (23.3% of total). While the absolute count is comparable to 2022 (26,814 claims), the proportional increase from 16.6% to 23.3% reflects the growing share of the overall risk portfolio this category occupies.

Sector	2023 MSK Claims	% of All MSK Claims
Health Care & Social Assistance	7,150	26.9%
Durable Goods Manufacturing	3,004	11.3%
Non-Durable Goods Manufacturing	2,589	9.7%
Construction	2,472	9.3%
Retail Trade	2,421	9.1%
Transportation & Warehousing	1,413	5.3%
All other sectors	7,575	28.4%

5.3 Machine-Related Injuries

Machine-related injury claims grew from 3,930 (2.4%) in 2022 to 4,605 (4.0%) in 2023 — a 17.2% increase in absolute terms and a significant proportional rise. This trend is concentrated in Manufacturing and Construction sectors and may reflect increased production volumes, staffing pressures, or reduced compliance with lockout/tagout safety procedures post-pandemic.

6. Recommendations

Based on the analysis, the following risk management priorities are recommended for organizational leaders and risk services teams:

Recommendation 1: Implement a Health Care Sector Risk Mitigation Strategy

Health Care and Social Assistance is the highest-risk sector across every dimension, physical injury, psychological harm, workplace violence, and musculoskeletal disorders. A dedicated sector risk strategy should include:

- Ergonomic assessments and patient handling protocol reviews to address overexertion and MSK claims.
- Workplace violence prevention training and incident reporting protocols, given the 2,173 assault claims in 2023.
- Psychosocial risk assessments and employee assistance program promotion to address the 2,352 psychological injury claims.
- Targeted monitoring for female workers aged 35-54, who carry the highest claim burden in this sector.

Recommendation 2: Escalate Workplace Violence as a Priority Risk

Assault and violence claims grew 28.4% year-over-year to 4,786 incidents in 2023. This is not a statistical artefact, it reflects a real and escalating hazard affecting Health Care, Education, and Public Administration. Recommended actions:

- Commission a workplace violence risk assessment for the top three affected sectors.
- Establish or review workplace harassment and violence prevention policies in alignment with applicable legislation.
- Introduce post-incident support pathways, as violence exposure is a primary driver of psychological injury claims.

Recommendation 3: Build a Proactive Musculoskeletal Injury Prevention Program

With 26,624 MSK claims representing 23.3% of all 2023 claims, and concentrated across both care and manual labour sectors, musculoskeletal prevention offers the highest volume reduction opportunity. Recommended actions:

- Introduce ergonomic risk screenings in high-volume MSK sectors (Health Care, Manufacturing, Construction, Retail).
- Establish return-to-work protocols that reduce re-injury risk for MSK claims.
- Track MSK claims as a standalone KPI in organizational risk dashboards.

Recommendation 4: Develop a Psychological Safety Monitoring Framework

The near-doubling of psychological injury claim share in a single year demands a structured response. Recommended actions:

- Designate psychological injury as a tracked risk category in annual risk reporting.
- Establish baseline psychological safety metrics (e.g., Employee Assistance Program utilization, absenteeism rates, stress-related leave) to triangulate CNESST claim data with internal indicators.
- Pilot psychological safety audits in Health Care and Education, the two sectors with the highest concentration.

7. Data Notes & Methodology

This analysis is based on the CNESST (Commission des normes, de l'équité, de la santé et de la sécurité du travail) Occupational Injuries open dataset for 2022 and 2023, obtained from the Quebec Open Data Portal (donneesquebec.ca). All data is publicly available and contains no personally identifiable information.

Dataset Summary

Parameter	Detail
Source	CNESST — Government of Quebec Open Data
Years covered	2022 and 2023
Total records analyzed	276,307 injury claims
2022 records	161,962
2023 records	114,345
Variables used	Injury nature, body part, incident type, causal agent, sex, age group, industry sector, 5 special risk flags
Language	Original data in French; translated to English for this report
Tools	Power BI & Power Query
Limitations	Data includes only accepted CNESST claims, minor injuries and unreported incidents are excluded. Year-over-year comparisons are affected by the post-pandemic normalization of COVID-19 claims.

This report was prepared as a work sample demonstrating skills in data analysis, KPI development, risk narrative construction, and executive-ready reporting. The analytical approach, insights, and recommendations are original and based solely on the public dataset described above.